



ENGINEERING DESIGN BRIEF

MANDATORY SPECIFICATION: PRODUCTION-GRADE EMBEDDED PLATFORM

This system must be implemented as a production-grade embedded platform, not a prototype.

The design must follow:

- Single binary per device (A and B)
- Internally modular architecture
- Strict event-driven communication
- Hardware-aware development (via SSH on Device A)

Core Principles:

- No tight coupling between modules
- Each subsystem must be independently testable
- Future upgrades must not require architectural redesign
- Debugging must be possible per module

PROJECT SPECIFICATION

Project Name: Larscope Embedded Medical Imaging System

Platform: Firefly CM3588 (Ubuntu Linux)

1. PROJECT CONTEXT AND OBJECTIVE

We are developing a medical-grade real-time imaging system ("Larscope") consisting of:

- A headless capture device (Device A)
- A monitoring and control device (Device B)

The system must support:

- High-quality video capture (4K, scalable to 8K)
- Real-time streaming (≤ 80 ms latency)
- Hardware control (LEDs, buttons, thermal)
- Reliable storage and synchronization

- Long-term scalability

2. DEVELOPMENT WORKFLOW (CRITICAL)

Development must be performed directly on Device A via SSH.

Required Process:

1. Inspect hardware: /dev/video*, GPIO chips, I2C buses
2. Validate drivers
3. Build and test modules independently
4. Integrate using event system
5. Finalize and push to GitHub

3. SYSTEM ARCHITECTURE OVERVIEW

Device A — Larscope (Headless Unit)

Primary responsibilities: Camera capture, LED control, Button handling, Recording (SD card), Streaming, Thermal monitoring, Configuration execution.

Device B — Monitoring & Control Unit

Primary responsibilities: Live video display, File access (from Device A), Configuration UI, Command transmission, System monitoring.

4. DEVICE A — DETAILED IMPLEMENTATION REQUIREMENTS

4.1 Core Behavior

On boot: Auto-start application; Initialize all modules; Enable camera + auto white balance; Turn ON illumination LEDs (default); Set status LED (Power ON); Start streaming service; Detect network (Ethernet/WiFi).

4.2 Camera Module

- Interface: USB (Sony IMX)
- Resolution: 4K (future 8K ready)
- Output: Live stream, Recording (H.264 / H.265), Still images (JPEG)

4.3 GPIO Input Module (Buttons)

Button	GPIO	Action
Middle	GPIO1_D6	Capture image

Button	GPIO	Action
Right	GPIO1_B3	LED Group 1
Left	GPIO1_B2	LED Group 2
Upper	GPIO1_B1	Zoom IN
Lower	GPIO1_B0	Zoom OUT

Requirements: Debounce (20–50 ms), Event generation only (no direct control).

4.4 Status LED Controller (GPIO)

LED	GPIO	Function
Power	GPIO3_A4	System ON
Battery	GPIO3_A3	Battery level
Charging	GPIO3_A2	Red/Green
WiFi	GPIO3_A1	Status
Recording	GPIO3_A0	Recording state

4.5 Illumination LED Controller (I2C)

- Drivers: 2 × TLC59108IRGYR
- Groups: Group 1 (8 LEDs), Group 2 (8 LEDs)
- Control GPIOs: GPIO0_C5, GPIO0_C4, GPIO0_D5, GPIO0_D4
- Behavior: Brightness cycle (75% → 50% → OFF), Smooth PWM transitions

4.7 Streaming Module

```
{
  "resolution": "3840x2160",
  "codec": "h265",
  "fps": 30,
  "latency_ms": 70
}
```

Requirements: RTSP default, ≤ 80 ms latency, Hardware encoding.

4.10 Thermal Manager

- Monitor SoC temperature
- Ensure external $\leq 40^{\circ}\text{C}$
- Trigger: LED dimming, Performance reduction

4.11 Event System (CORE)

All modules communicate via events:

```
{  
  "event": "button_press",  
  "button": "right",  
  "gpio": "GPIO1_B3"  
}
```

5. DEVICE B — DETAILED IMPLEMENTATION REQUIREMENTS

- **5.2 Video Module:** Connect to Device A stream, Display low-latency video, Zoom visualization.
- **5.3 File Sync Module:** Fetch files from Device A (incremental/resume support).
- **5.4 Configuration UI:** Protocol selection, Codec, Resolution, Network mode.

6. SOFTWARE ARCHITECTURE

Device A Modules: GPIO Input, Status LED, Illumination LED, Camera, Recording, Streaming, Network, Config, Thermal, Event Dispatcher.

Device B Modules: UI Module, Stream Client, Sync Manager, Config Manager, Command Client.

7. GITHUB STRUCTURE

```
larscope/  
├─ device_a/  
├─ device_b/  
├─ shared/  
└─ docs/
```

8. RELIABILITY REQUIREMENTS

- Headless operation (Device A)
- Auto-start on boot
- Crash recovery
- Per-module logging
- Fault isolation

FINAL REQUIREMENT: *The system must be Modular, Debuggable, Scalable, Hardware-validated, and Production-ready.*
END OF PROMPT